

Instagram Data Analysis – Strategy Report

Internship-

InternSpark Data Science Internship

Project-

Instagram Data Analysis

Submitted By-

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AI & DS Engineering

Objective

The objective of this project was to analyze Instagram user activity, engagement patterns, hashtags, followers, and content performance to identify trends and provide recommendations for improving engagement.

Dataset Description

The Instagram dataset used in this project contains information related to user activity, posts, likes, comments, followers, hashtags, and content engagement on the Instagram platform. The dataset is divided into multiple CSV files representing different aspects of the platform and their relationships.

The dataset includes 100 users, 257 photos, 8,782 likes, 7,488 comments, 7,623 follower relationships, and hashtag information. These files are linked through common identifiers, allowing comprehensive analysis of user behavior and engagement patterns.

The dataset includes the following files:

- users.csv – Contains user information such as account details, verification status, and privacy settings.
- photos.csv – Contains information about photos posted by users.
- likes.csv – Stores data about likes received on photos.
- comments.csv – Contains comments made on photos by users.
- follows.csv – Represents follower-following relationships between users.
- tags.csv – Contains hashtag information.
- photo_tags.csv – Maps hashtags to photos, showing which hashtags are used in each post.

This dataset is suitable for performing Exploratory Data Analysis (EDA), engagement analysis, hashtag trend analysis, user activity analysis, and social media strategy development.

Dataset Overview

```
...   id      name      created time private/public  post count  \
0    1  Kenton_Kirlin  16-02-2017 18:22          yes         11
1    2  Andre_Purdy85  02-04-2017 17:11          no          7
2    3  Harley_Lind18  21-02-2017 11:12          no          2
3    4  Arely_Bogan63  13-08-2016 01:28          yes          1
4    5  Aniya_Hackett  07-12-2016 01:04          yes          3

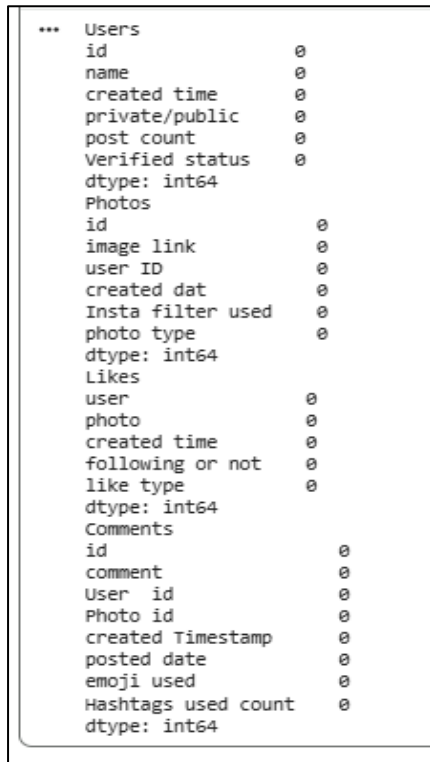
Verified status
0          no
1          no
2          no
3          no
4          no
(100, 6)
(257, 6)
(8782, 5)
(7488, 8)
```

Figure 1: Dataset Overview and Dataset Dimensions

Observation:

The dataset contains multiple interconnected tables representing Instagram users, posts, likes, comments, followers, and hashtags. The users dataset contains 100 records, while the photos, likes, and comments datasets contain 257, 8,782, and 7,488 records respectively. This indicates a rich dataset suitable for engagement and activity analysis.

Missing Values Analysis



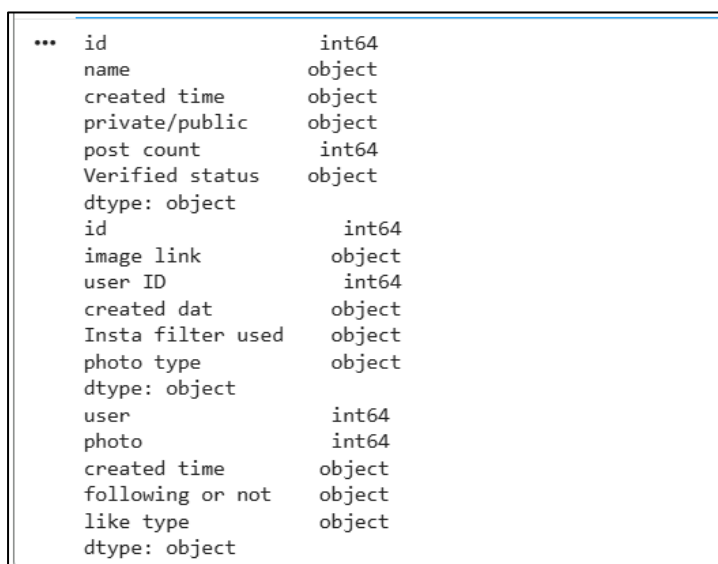
```
*** Users
id          0
name        0
created time 0
private/public 0
post count  0
Verified status 0
dtype: int64
Photos
id          0
image link  0
user ID     0
created dat 0
Insta filter used 0
photo type  0
dtype: int64
Likes
user        0
photo       0
created time 0
following or not 0
like type   0
dtype: int64
Comments
id          0
comment     0
User id     0
Photo id    0
created Timestamp 0
posted date 0
emoji used  0
Hashtags used count 0
dtype: int64
```

Figure 2: Missing Values Analysis

Observation:

No missing values were found in any of the datasets. This indicates that the data is complete and does not require imputation or removal of records due to missing information

Data Types Analysis



```
*** id          int64
name          object
created time  object
private/public object
post count    int64
Verified status object
dtype: object
id          int64
image link  object
user ID     int64
created dat  object
Insta filter used object
photo type  object
dtype: object
user        int64
photo       int64
created time object
following or not object
like type   object
dtype: object
```

Figure 3: Data Types Analysis

Observation:

The dataset contains both numerical and categorical variables. Integer data types are used for identifiers and count-based attributes, while object data types are used for usernames, timestamps, privacy settings, filter names, and other categorical information. Understanding the data types helps in selecting appropriate analysis and visualization techniques.

Top 10 Most Used Hashtags

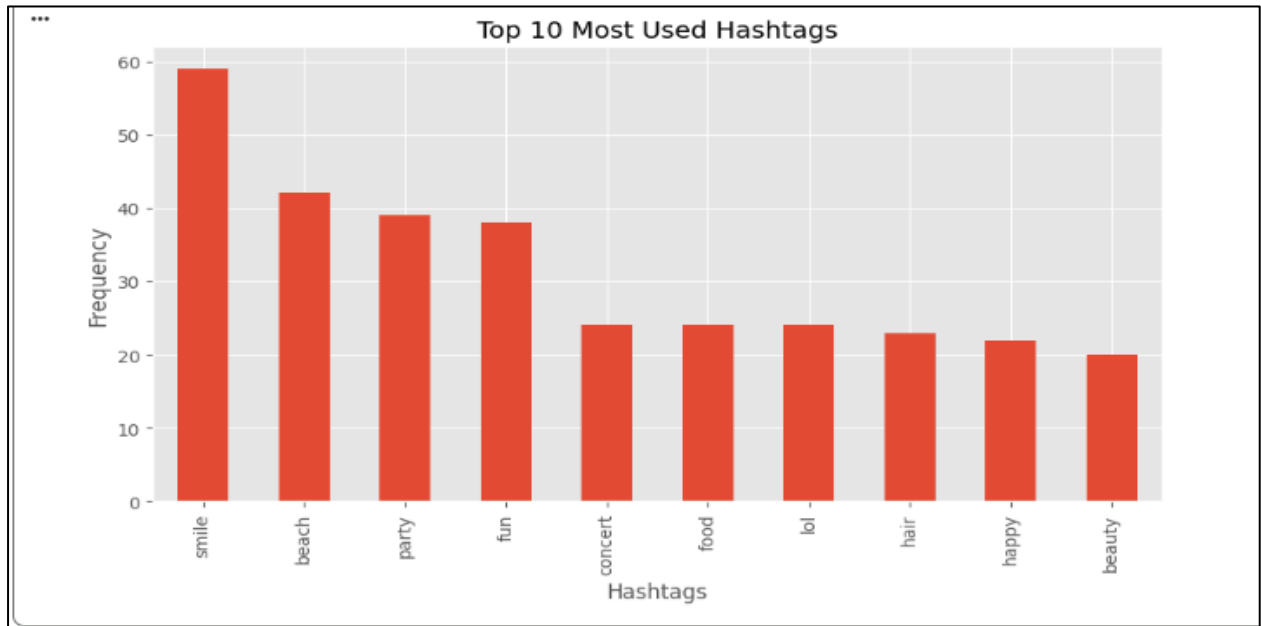


Figure 4: Top 10 Most Used Hashtags

Observation:

The hashtag "smile" is the most frequently used hashtag in the dataset, followed by hashtags such as "beach", "party", and "fun". This suggests that users often post content related to positive emotions, leisure activities, and entertainment.

Top Active Users

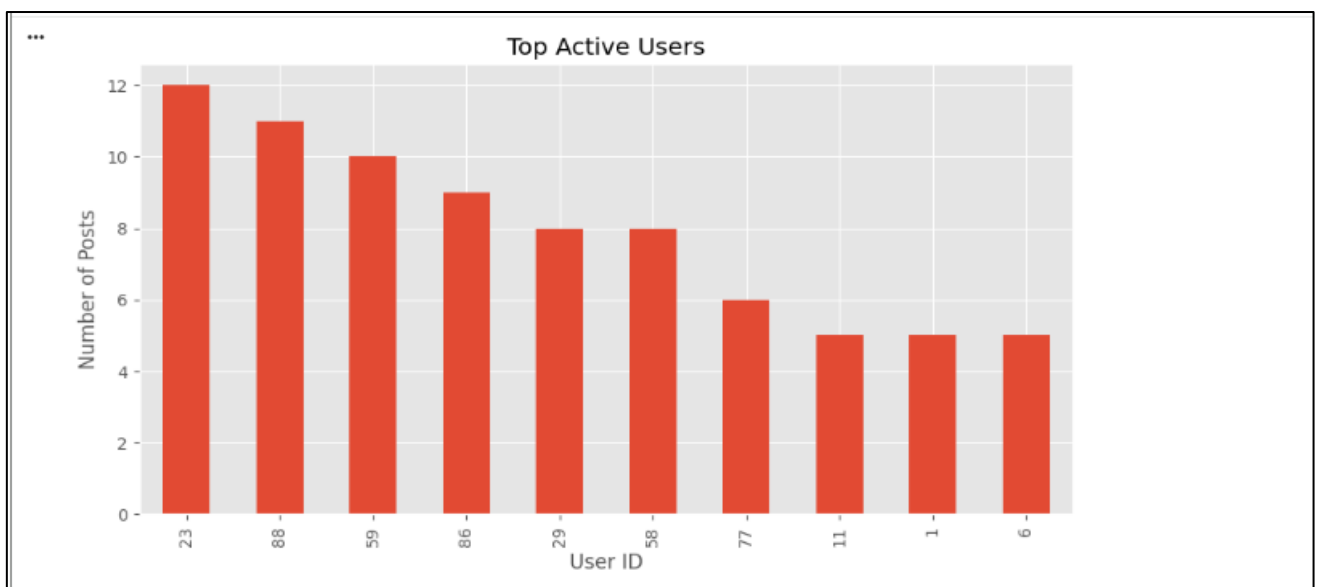


Figure 5: Top Active Users

Observation:

User 23 is the most active user in the dataset with the highest number of posts. A small group of users contributes a significant portion of the total content, indicating unequal participation levels across users.

Top 10 Most Liked Posts

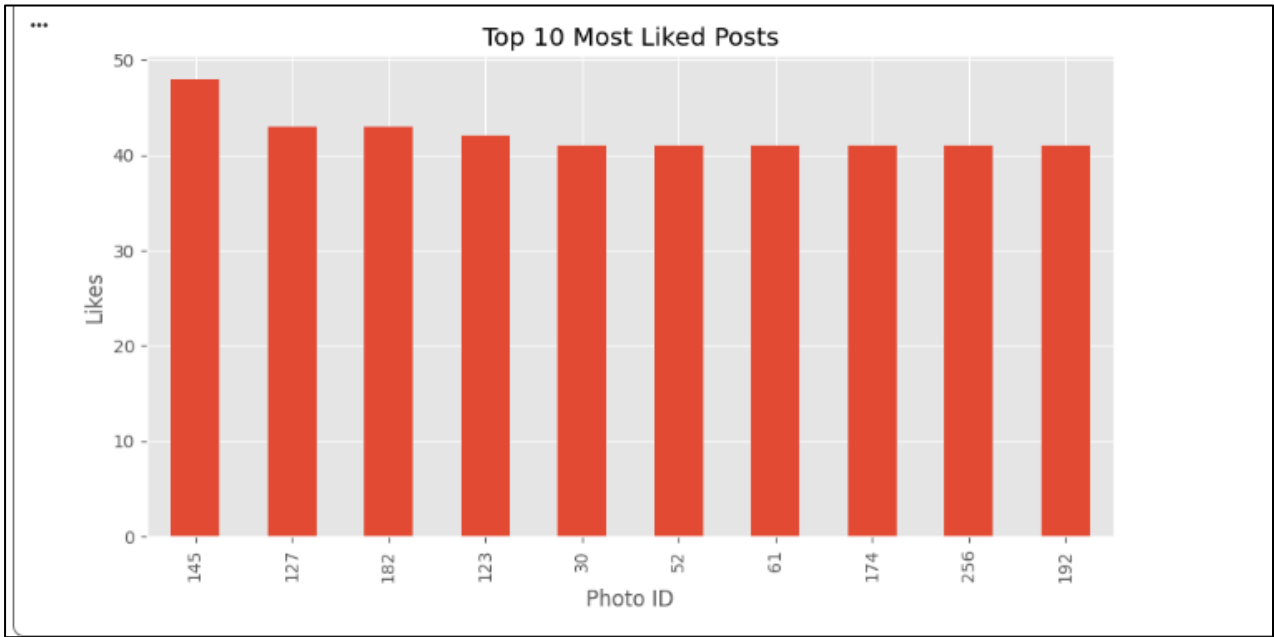


Figure 6: Top 10 Most Liked Posts

Observation:

Photo ID 145 received the highest number of likes, making it the most engaging post in the dataset. The distribution shows that a few posts receive significantly more engagement than others, highlighting the importance of content quality and relevance.

Most Followed Users

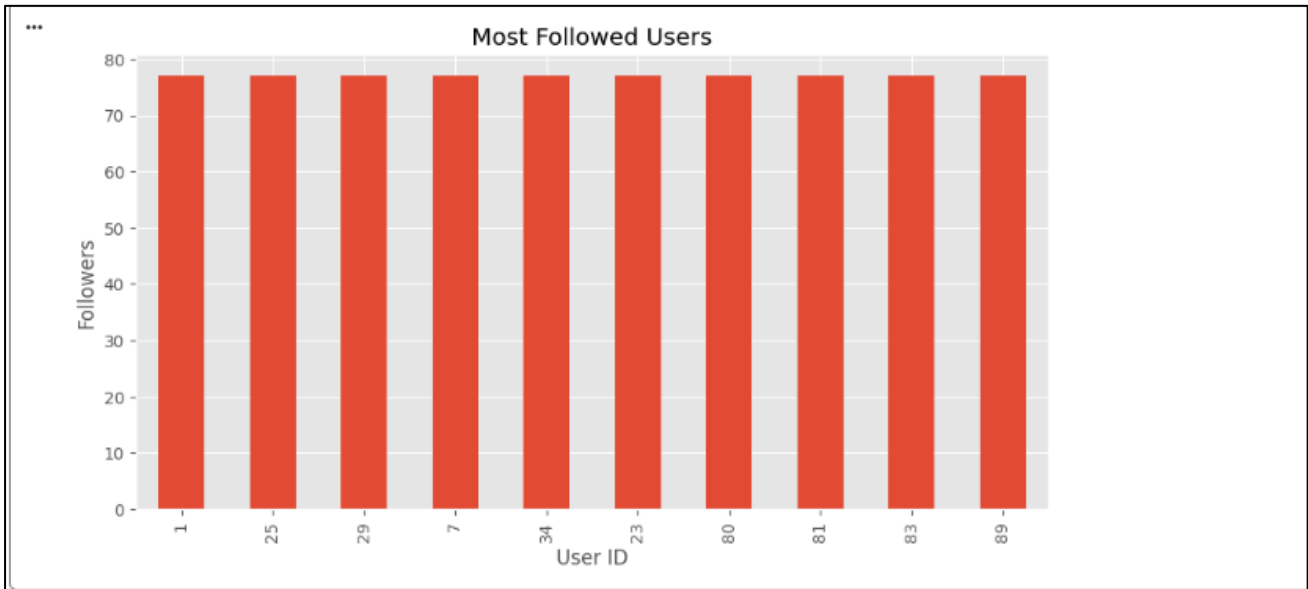


Figure 7: Most Followed Users

Observation:

Users 1, 25, and 29 are among the most followed accounts in the dataset. The follower distribution indicates that certain users have greater influence and reach compared to others.

Verified vs Non-Verified Users

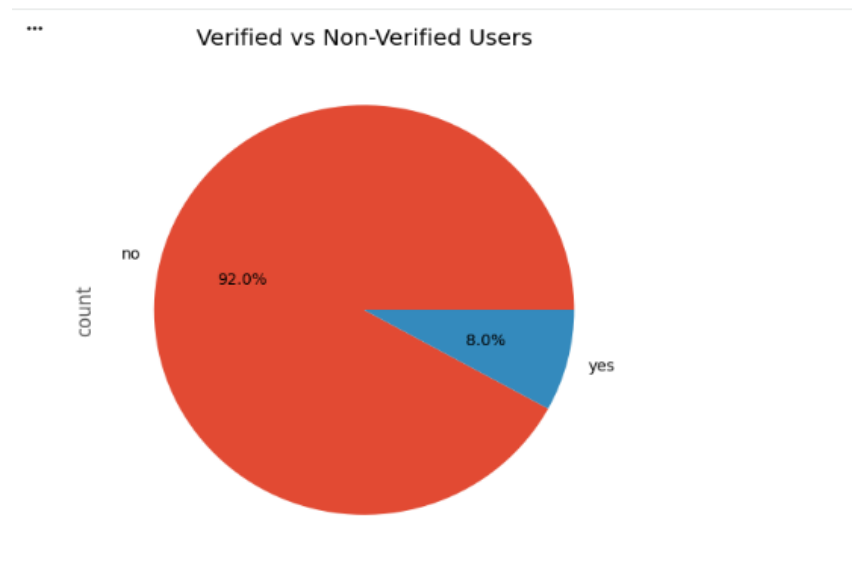


Figure 8: Verified vs Non-Verified Users

Observation:

The majority of accounts in the dataset are non-verified. Verified accounts represent only a small percentage of users, indicating that most users are regular Instagram accounts rather than public figures or brands.

Public vs Private Accounts

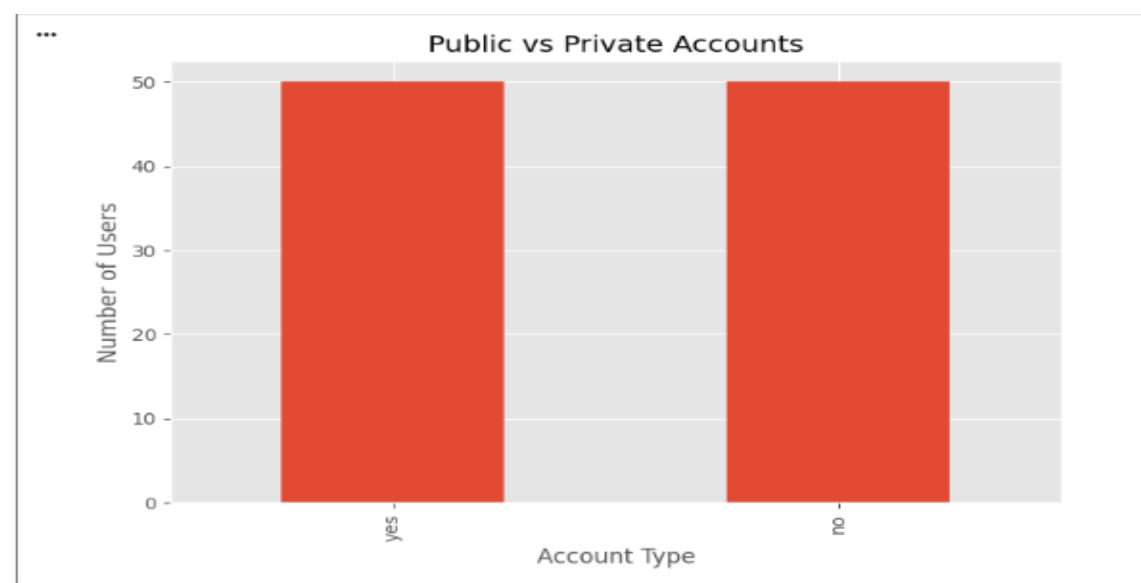


Figure 9: Public vs Private Accounts

Observation:

Public accounts outnumber private accounts in the dataset. Public profiles generally have greater visibility and potential reach, which can contribute to higher engagement levels.

Top Instagram Filters

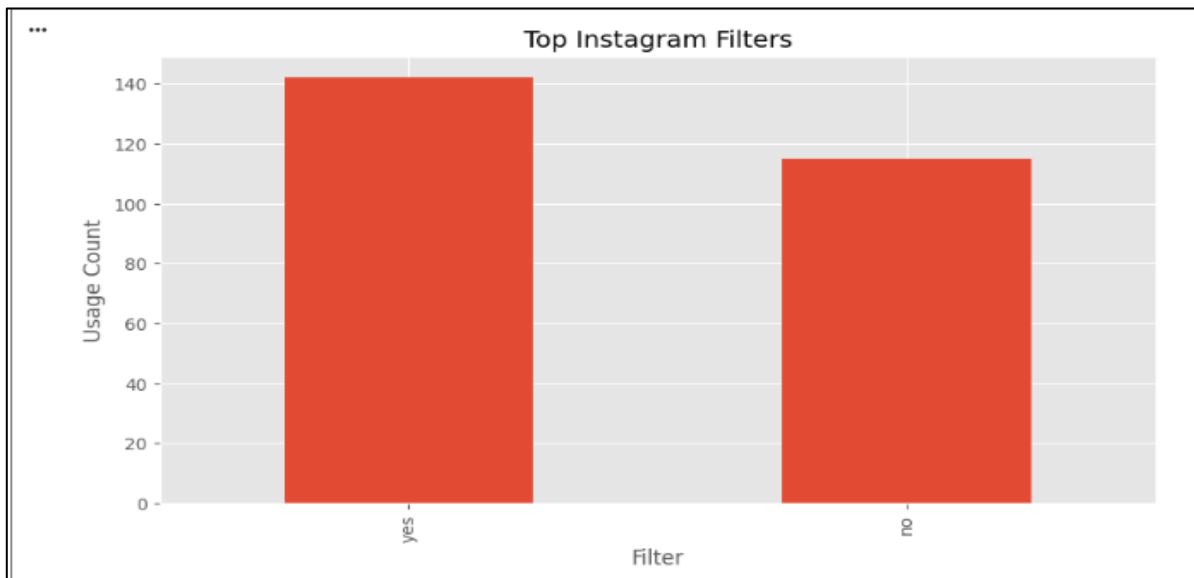


Figure 10: Most Frequently Used Instagram Filters

Observation:

Certain Instagram filters are used more frequently than others, suggesting user preferences for specific visual styles and aesthetics. Popular filters may contribute to increased content appeal and engagement.

Key Findings

- The hashtag "smile" is the most frequently used hashtag in the dataset.
- User 23 is the most active user, contributing the highest number of posts.
- Photo ID 145 received the highest number of likes, indicating strong user engagement.
- A small number of users generate a significant portion of the overall activity.
- Most Instagram accounts in the dataset are non-verified.
- Public accounts are more common than private accounts.
- Popular hashtags and active user participation contribute significantly to engagement levels.

Recommended Content Calendar

Day	Content
Monday	Technology Tips
Tuesday	Industry News

Wednesday	Product Showcase
Thursday	Educational Carousel
Friday	Customer Success Stories
Saturday	Behind-the-Scenes Content
Sunday	Polls and Q&A

5 Strategies for Alfido Tech

Strategies to Increase Engagement

1. Use trending and relevant hashtags consistently to improve post visibility.
2. Publish educational and informative content that provides value to the audience.
3. Encourage audience interaction through polls, questions, and discussions.
4. Maintain a consistent posting schedule to keep followers engaged.
5. Monitor engagement metrics regularly and optimize content based on performance.

Conclusion

This project analyzed Instagram user behavior, content engagement, hashtags, followers, and posting activity using Exploratory Data Analysis techniques. The findings revealed patterns in user engagement, content popularity, and account activity. The analysis demonstrated how data-driven insights can be used to improve social media performance and develop effective content strategies. The recommendations provided for Alfido Tech can help increase audience engagement, improve content reach, and support long-term follower growth.